

Objective

The motive of this research is to understand the effects of ozone concentration in the atmosphere to the spread of coronavirus infection rates in different regions of the United States of America.

Ozone, as we know, protects us from the harmful UV radiation from the Sun and prevents us from developing diseases like skin cancer, skin-ageing, etc. This protection is done a few miles above the surface level. But ozone near the ground level where we live and breathe has an opposite effect altogether. It is shown that breathing ozone directly can lead to weakening of the human immune system, making us more prone to catch infection like the coronavirus with ease. It is important we understand the effects of ozone to a greater extent as high concentrations of ozone near the surface are not normal as the chemical reactions required to produce it are accelerated heavily by human activities.

This is performed for the year of 2022, when coronavirus was at a peak. The regions chosen for this study were picked based on varying climatic conditions and with a very high population density.

Hypothesis

The correlation between ozone and coronavirus depends on the climatic conditions of the region and the overall population density of the region. Some regions may not show any correlation between the two because of differences in average temperature levels or due to the presence of other pollutants in the air.

Background

Coronavirus outbreak, as we know, forced the whole world to go remote for two good years. Understanding the factors that lead to the spread of such infection is important. Ozone around us could potentially be one of those catalysts that aid largely spreading the infection. Hence, a greater knowledge on the correlation between pollutants like ozone and coronavirus infection holds a lot of promise to tackle medically related diseases in the future.

There have been a handful of papers based on this topic. One such paper (Zoran et al., 2020) found that there is a correlation of 0.74 b/w coronavirus and ozone which overwhelmingly is a very high positive correlation. The study was performed in Milan, Italy during the early outbreak of the virus in the first half of 2020. Other pollutants and environmental factors such as relative humidity and air temperature were also correlated with the infection (figure 1).

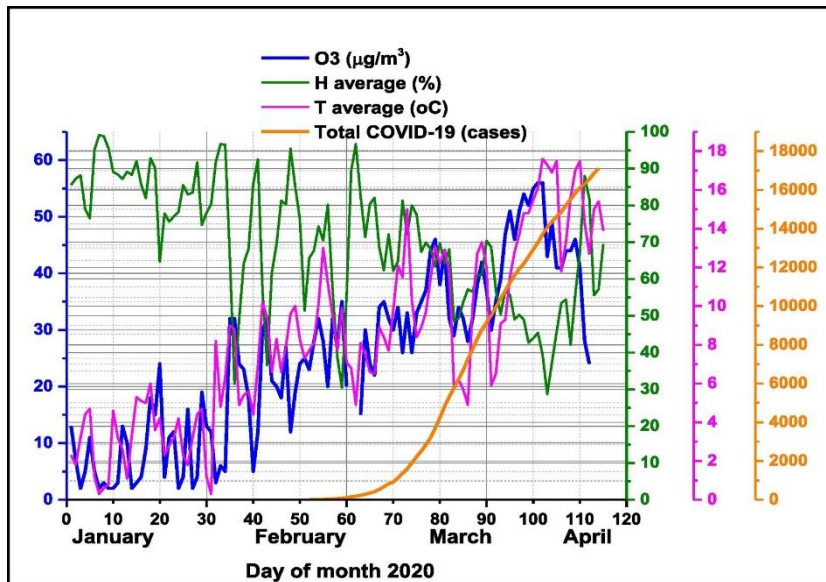


Figure 1: Zoran et al., 2020. Shows a positive correlation b/w total covid cases (yellow) and ozone (blue), and temperature (yellow). A negative correlation is seen b/w covid cases and relative humidity (blue).

Data Sources

The required data for this study was collected from websites handled by the United States government. The ozone data used is linked [here](#). The covid data used is linked [here](#). The data used includes information of ozone and coronavirus by counties for the year of 2022. Ozone is measured in ppm (parts per million) concentration in the air and covid cases reported weekly in each county in America.

The coronavirus data had a technical issue with the dates included as the column header instead of being present in a row. This was fixed by transposing the data by using a simple script written in python. Another issue that was encountered was that not every county had ozone concentration measured for (figure 4). There are a total of 3143 counties in the United States of which only 2800 counties had their ozone levels measured. However, this was not a major issue, as the study was done on a county scale.

Methods

3 counties spread across the USA were chosen based on a highly dense population and varying temperature conditions (figure 2 and 3). These 3 counties include:

1. Marion County, Indiana
2. Los Angeles, California
3. Miami Dade, Florida

USA Political Map with Highlighted States Used for this Study

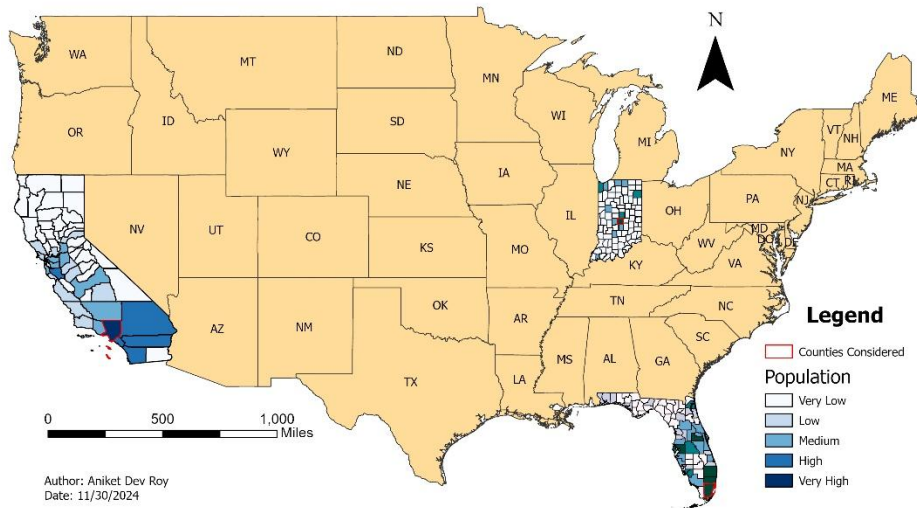


Figure 2: USA Political map. Highlighted states include California, Indiana, and Florida used in this study. They are symbolized based on population.

USA Annual Average Temperature Map

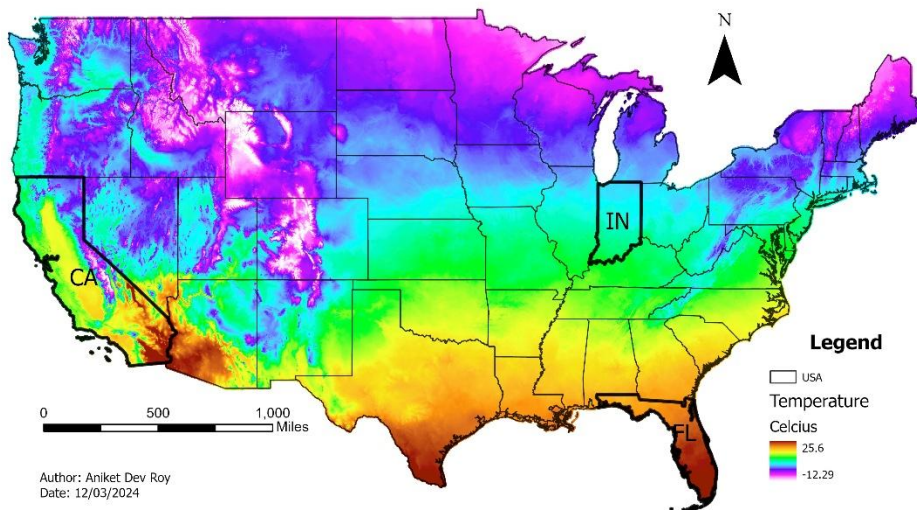
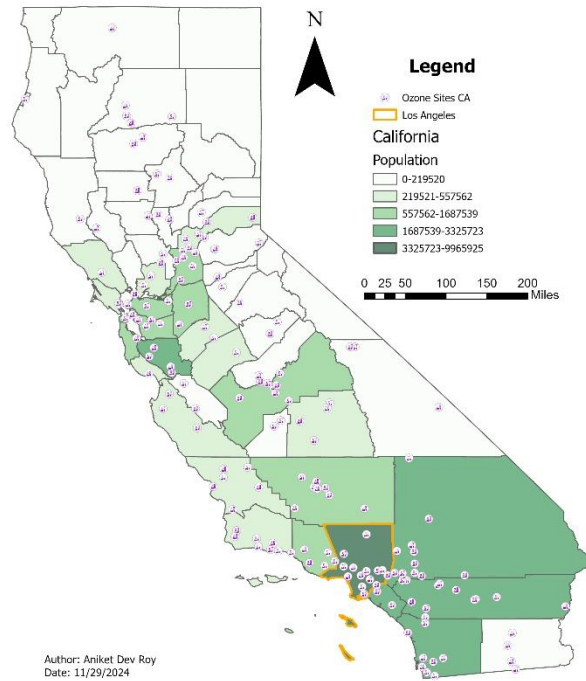
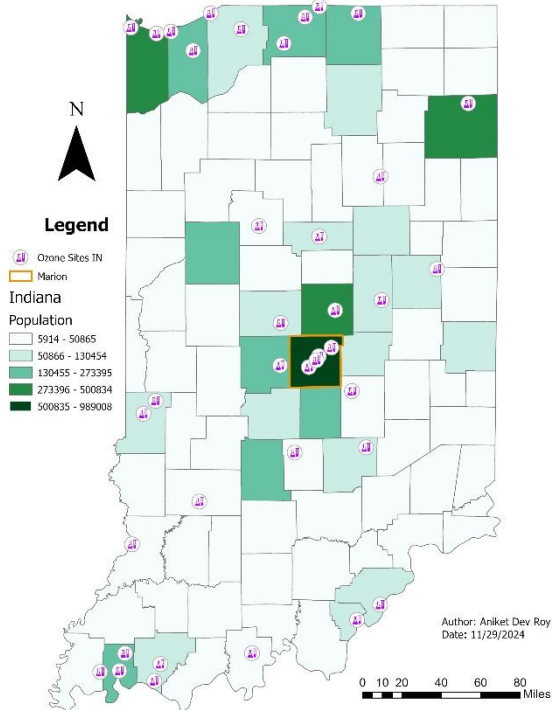


Figure 3: Annual average temperature USA map. CA, IN, FL state borders are bolded. All 3 states have varying temperatures.

All the 3 counties chosen had more than one air quality measurement site that found the ozone concentration in the air locally. The ozone and coronavirus data, for the year of 2022, were combined for each county and a **scatter plot** was made and finally, the correlation value between the two was found.

Indiana Counties Population and Ozone Sites California Counties Population and Ozone Sites



Florida Counties Population and Ozone Sites

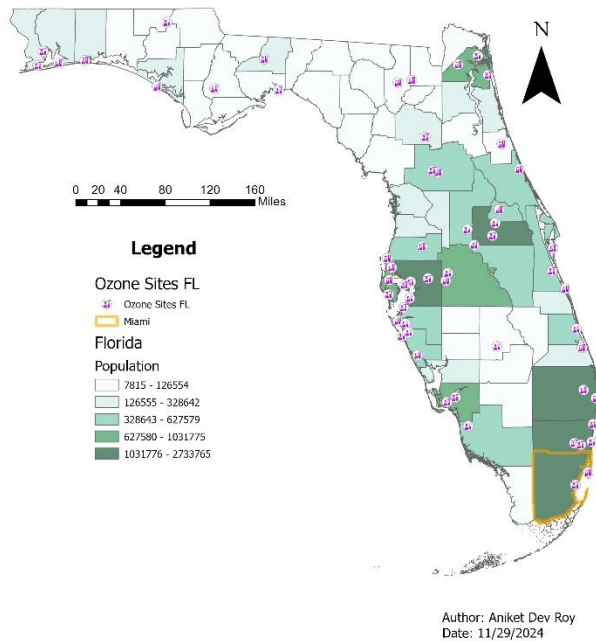
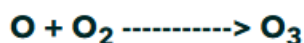
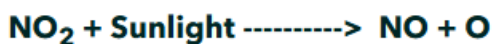


Figure 4: Indiana, California, and Florida state maps symbolized based on population. Counties considered in this study are highlighted in gold. Notice not all counties have ozone measurement sites.

The first 6 months of 2022 were considered to obtain the correlation. This is due to the increase in solar radiation from the Sun reaching the surface as we transitioned from a colder January to a relatively hotter June.

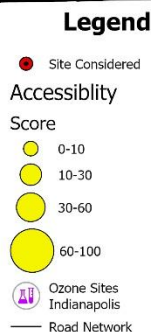
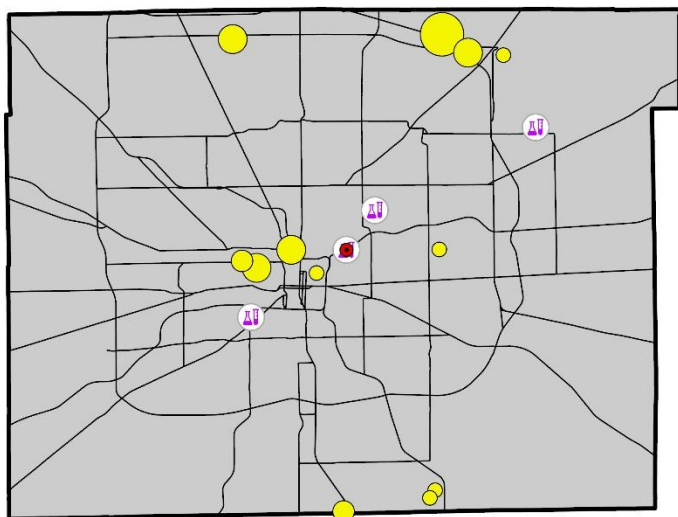
Sunlight plays a crucial role in the production of single oxygen atoms that further react with an oxygen molecule to produce ozone near the ground. Note that NO_x and VOCs are compounds released both by human induced activities, like fuel combustion, and naturally from the soil.



Discussion

Three different counties produced three different results. This was expected. An exponential rise in covid cases was seen in all 3 counties between January and March of 2022.

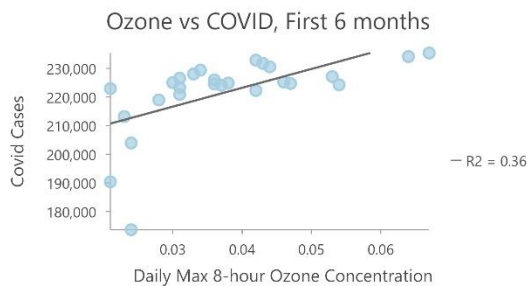
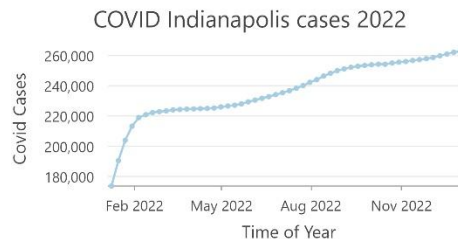
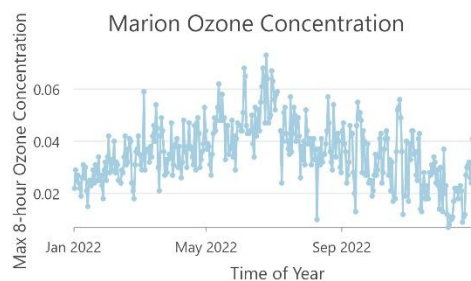
COVID Cases vs Ozone in Indianapolis, 2022



0 5.5 11 Miles

25% of Indianapolis effected by COVID in 2022.

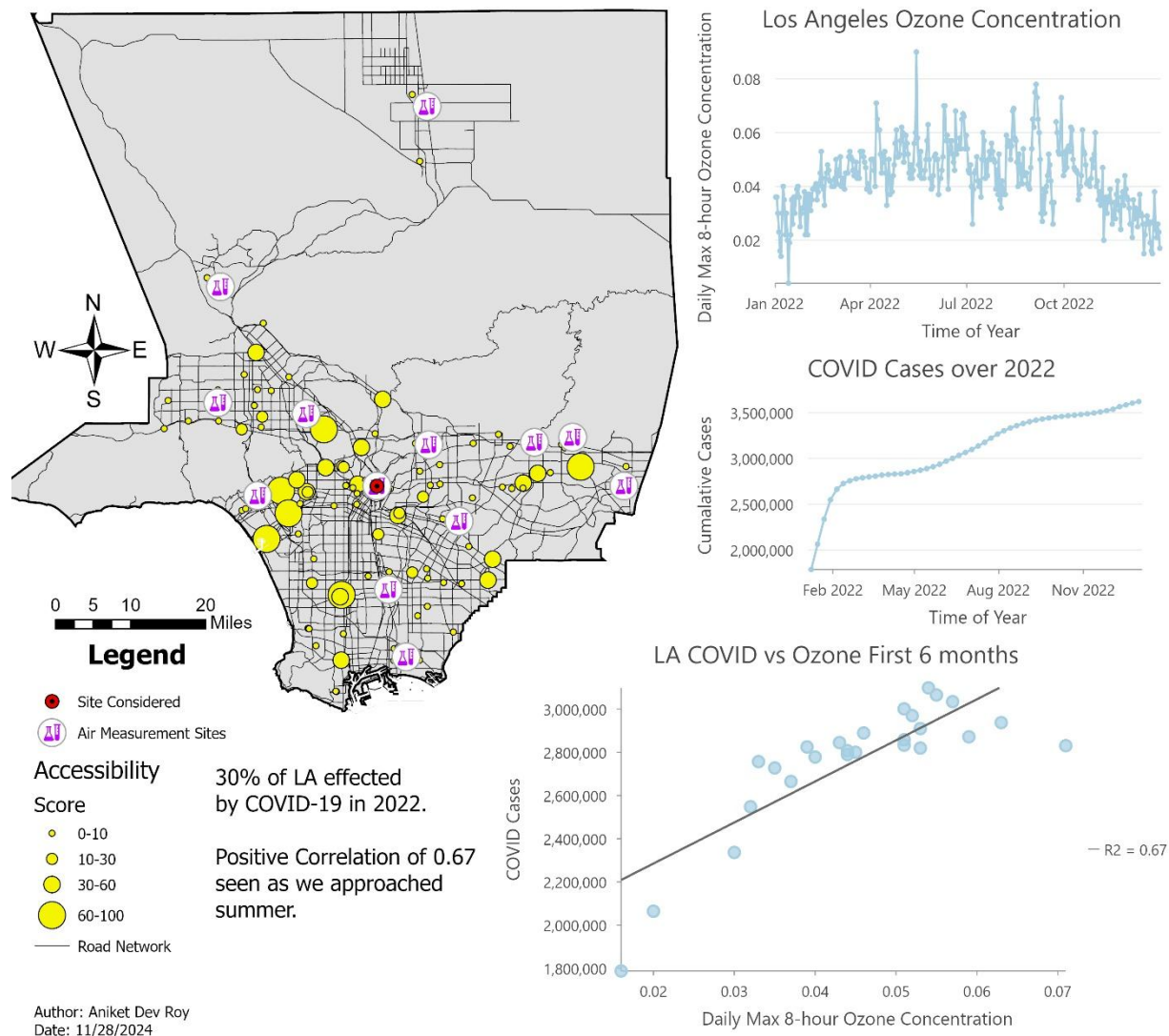
Not a significant Positive Correlation seen as we approach warmer months.



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Firstly, Marion County provided with a correlation between ozone and coronavirus of 0.36. This value is not high enough to definitively say that ozone is responsible for the spread of coronavirus in this county. By 2022, around 25% of the population in the county were affected which is a quarter and a significant portion of people living in that area.

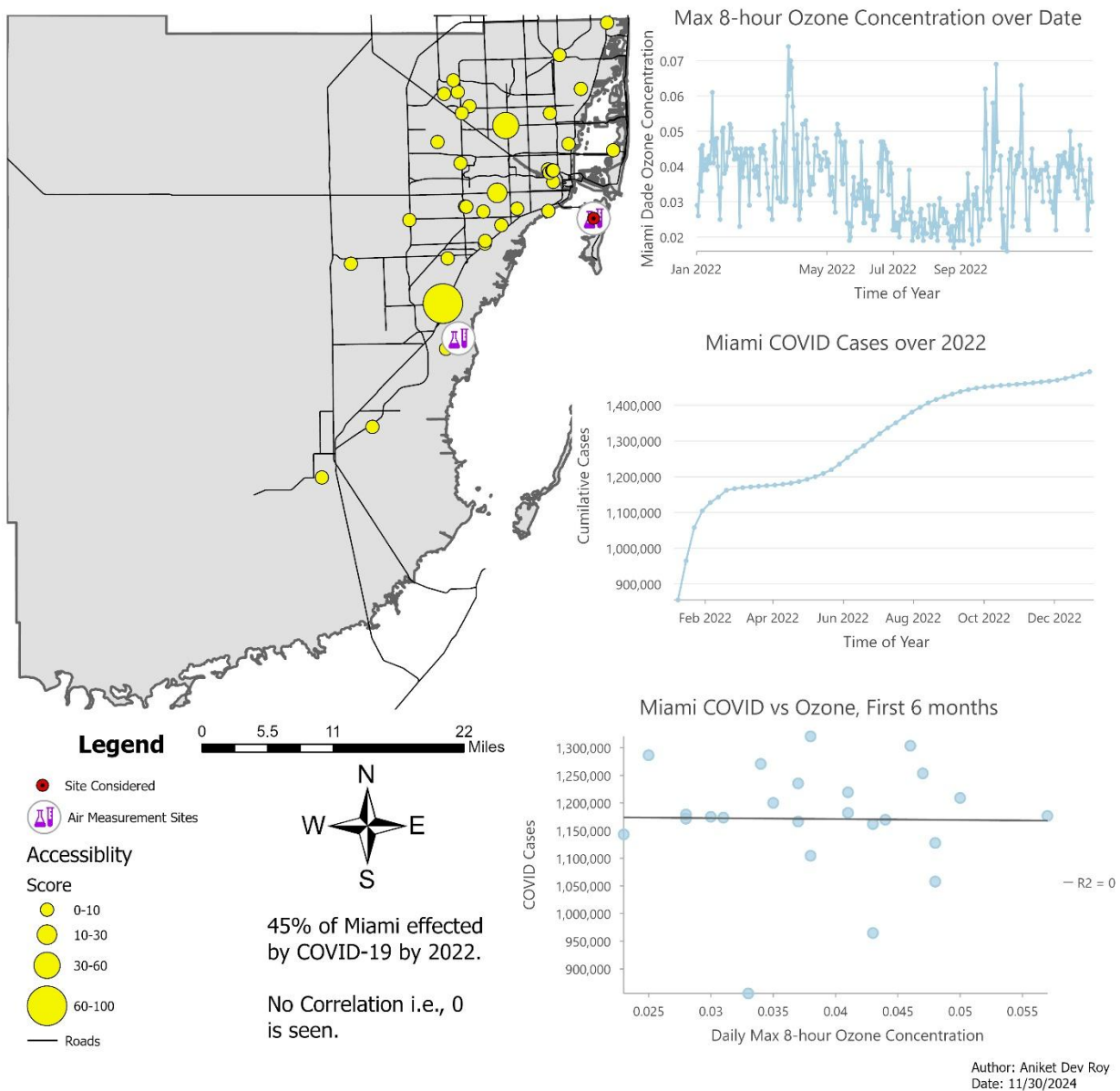
COVID Cases vs Ozone in Los Angeles, 2022



Secondly, Los Angeles provided a correlation of 0.67. This is a high positive correlation and hence a conclusion could perhaps be drawn that ozone could likely be the cause of spread of the coronavirus in this area. About 30% of the people living in LA were affected by the infection which is around 1/3rd of the population in the area.

Finally, Miami showed no correlation between ozone and coronavirus, giving a value of 0. Looking closely at the ozone plot it is understood that unlike Marion and LA where an increase in ozone concentration is seen in the summer, for Miami it is the opposite where a decrease in concentration in ozone is seen in the summer. This is likely due to Miami having a relatively warmer winter than the other two counties, and on average having a higher temperature throughout the year since it is present really close to the equator. Almost 45% of this county was affected by coronavirus in 2022.

COVID Vs Ozone in Miami-Dade, 2022



For each county, medical facility accessibility scores were also provided through road networks in the area. This directly correlates how controllable the infection was during its peak in 2022. Los Angeles had the highest number of acute care hospitals and even though it had a population over 10 million people at the time, around 1/3rd of the population was affected by the coronavirus. This could perhaps be related to an accelerated spread of infection by a factor unknown to us or it could be covid. A correlation of 0.67 was found for Los Angeles.

Conclusion

A conclusion can be drawn from the combined results of these regions that ozone is not likely the cause of exponential spread of the coronavirus in the first half of 2022. At least in Miami and Marion counties. As for comparing it with the results of Zoran et al, 2020, they performed the study for only one region and got a correlation of 0.74. This eliminates the factor of varying climatic conditions and other factors such as a different pollutant that could likely be a cause for the rise in the infection rates in a particular region. Finally, this idea of understanding comparisons between two completely unrelated variables, in this case ozone and coronavirus, needs further attention to be researched more as there is a lot grey area currently in this field that need to be resolved.

GIS Skills Summary

Quite a few GIS skills were applied in this study.

1. Transposing of the data feature. It was attempted in GIS, but the process was tedious so, ended up being performed in python.
2. Attribute selection. This was done for both coronavirus and ozone datasets to extract the data just for the year of 2022 and counties that had their ozone measured.
3. Join and Relate function. This was used to combine both the datasets based on the common date the data was recorded at.
4. Pairwise clipping. To remove any excess data after it was visualized. This was done for the road networks in interest.
5. Spatial join, Erase, and Buffer. This was used to understand medical accessibility to parts of the counties selected as to deduce how controllable the coronavirus was.
6. Copy features. This was used mainly to use data that was only readable and not editable.
7. Other skills include mapping through layout and required data search valid to complete the project.